```
royhills/arp-scan)
10.0.0.1   2c:54:91:88:c9:e3  (Unknown)
10.0.0.100 8c:1f:64:a2:d0:01  IEEE Registration Authority
10.0.0.103 00:0a:95:9d:68:16  (Unknown: locally administered)

3 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.9.7: 3 hosts scanned in 0.734 seconds (4.09
hosts/sec). 3 responded
```

Options

The following options can be used for the outgoing ARP packets:

Option	Default	Comments
–arphrd	1	Hardware type
–arppro	0x0800	Protocol type
–arphln	6	Hardware address length
–arppln	4	Protocol address length
–arpop	1	Opcode
–arpsha	Hardware address	Sender hardware address
–arpspa	IP address	Sender network protocol address
–arptha	0	Target hardware address

An example for the hardware address length option is shown below:

```
$ sudo arp-scan --arphln=6 -l
Interface: eth0, type: EN10MB, MAC: d4:ba:ba:af:ff:f1, IPv4:
10.0.0.103
Starting arp-scan 1.9.7 with 256 hosts (https://github.com/
royhills/arp-scan)
10.0.0.1 2c:54:91:88:c9:e3 (Unknown)
10.0.0.102 8c:1f:64:a2:d0:01 IEEE Registration Authority
10.0.0.1 2c:54:91:88:c9:e3 (Unknown) (DUP: 2)
10.0.0.1 2c:54:91:88:c9:e3 (Unknown) (DUP: 3)
10.0.0.1 2c:54:91:88:c9:e3 (Unknown) (DUP: 4)

5 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.9.7: 256 hosts scanned in 1.973 seconds
(129.75 hosts/sec). 5 responded
```

The Ethernet frame options include:

Option	Default	Comments
–destaddr	ff:ff:ff:ff:ff:ff	Destination address
–srcaddr	Interface address	Source address
–prototype	0x0806	Protocol type

Format

The header and footer text in the output can be suppressed by using the --*plain* option as shown below:

```
$ sudo arp-scan -l --plain
10.0.0.1   2c:54:91:88:c9:e3  (Unknown)
10.0.0.100 8c:1f:64:a2:d0:01  IEEE Registration Authority
```

In the *arp-scan* 1.10.0 version, you have the --*format* option for the output:

```
$ sudo arp-scan --plain --format='${ip},${mac},"${vendor}"' -l
10.0.122.1,52:54:00:37:45:95,"QEMU"
```

PCAP

The responses from *arp-scan* can be saved to a *.pcap* file using the --*pcapcsavefile* option. Here's an example:

```
$ sudo arp-scan -I eth0 --pcapsavefile=arp.pcap 10.0.0.100
Interface: eth0, type: EN10MB, MAC: d4:ba:ba:af:ff:f1, IPv4:
10.0.0.104
Starting arp-scan 1.9.7 with 1 hosts (https://github.com/
royhills/arp-scan)
10.0.0.100 8c:1f:64:a2:d0:01  IEEE Registration Authority

1 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.9.7: 1 hosts scanned in 0.173 seconds (5.78
hosts/sec). 1 responded
```

You can view the contents of the *.pcap* file using *tcpdump* as follows:

```
$ tcpdump -n -r arp.pcap
reading from file arp.pcap, link-type EN10MB (Ethernet),
snapshot length 64
20:40:04.125747 ARP, Reply 10.0.0.100 is-at
8c:1f:64:a2:d0:01, length 46
```

arp-fingerprint

The *arp-fingerprint* command can be used to fingerprint a specific host. You cannot provide multiple networks, or hosts as input. It uses *arp-scan* to send and receive the ARP requests. An example is shown below:

```
$ sudo arp-fingerprint -o "--interface=eth0 --numeric"
10.0.0.100
10.0.0.100 01010100000Linux 2.2, 2.4, 2.6, 3.2, 3.8, 4.0,
4.6, Vista, 2008, Windows7, Windows8, Windows10

$ sudo arp-fingerprint -o "--interface=wlan0 --numeric"
10.0.1.15
10.0.1.15  11110000000Linux 2.0, MacOS 10.4, IPSO 3.2.1,
Minix 3, Cisco VPN Concentrator 4.7, Catalyst 1900, BeOS,
WIZnet W5100
```

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